

# GABRIEL ASael TARIGAN

## Curriculum Vitae

Lecturer Specialist • Artificial Intelligence • Machine Learning

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## PROFILE

I am a Lecturer Specialist, Subject Developer, and Researcher in the Artificial Intelligence Program at BINUS University's School of Computer Science, where much of my work is driven by a simple belief: AI should not stay confined to those who already have access to it. Beyond developing curricula and researching deep learning, computer vision, time-series forecasting, and MLOps, I coordinate ARISE and AIPLEX, national AI competitions that bring machine learning to students, including many encountering it for the first time, through hands-on, Kaggle-style challenges. I also build the ML platforms, automated assessment systems, and interactive learning tools that make this kind of large-scale, hands-on AI education possible, and I extend the same effort into community outreach across Indonesia. I hold a Master of Computer Science from BINUS University (Summa Cum Laude, GPA 3.95) and AI certifications from Microsoft and Huawei.

## EDUCATION

### BINA NUSANTARA UNIVERSITY

Master's degree in computer science. **3.95/4.00, Summa Cum Laude**

Jakarta

2023–2024

### SOEGIJAPRANATA CATHOLIC UNIVERSITY

Bachelor's degree in computer science. **3.79/4.00, Best Graduate**

Semarang

2018–2022

## EXPERIENCE

### BINA NUSANTARA UNIVERSITY

Lecturer Specialist, Subject Developer & Researcher; Artificial Intelligence Program

Jakarta

2025–Present

- Teach undergraduate and graduate AI courses across the program, from machine learning foundations to MLOps.
- Author the program's core curriculum: 7 courses spanning ML, deep learning, applied NLP, computer vision, MLOps, signal processing, IoT and robotics, including course outlines, assessment design, and labs.
- Build and operate production platforms: a Kaggle-style ML competition platform (ARISE), automated assessment portals, and interactive learning tools.
- Coordinate national AI competitions for students and lead community outreach for AI literacy.

### EMACS

Peer Reviewer

Jakarta

2024–Present

Acted as a peer reviewer for the EMACS conference, evaluating submitted manuscripts for technical soundness, methodological rigor, originality, and relevance to the field. Provided constructive feedback to support the review process.

## ACTIVITIES

### ARTIFICIAL INTELLIGENCE RESEARCH AND INTELLIGENT SYSTEMS EVALUATION (ARISE)

Jakarta

Coordinator

2026

- Coordinate ARISE, a Kaggle-style ML competition for students.
- Built and operate a custom platform with live leaderboards and automated evaluation.

## **AIPLEX – AI PROTOTYPE AND LOGIC EXHIBITION, BEEFEST**

**Jakarta**

### **Coordinator**

**2025**

- Developed and coordinated AIPLEX end to end, a national AI competition for high school students.
- Themed the competition around AI for everyday problems, aligned with the SDGs.

## **ICCCSI CONFERENCE**

**Jakarta**

### **Session Chair**

**2024-2025**

- Served as session chair for the International Conference on Computer Science and Computational Intelligence.
- Handled opening duties for the conference.

## **COMMUNITY OUTREACH**

**Indonesia**

### **Volunteer**

**Present**

- Deliver AI and programming workshops for schools: SMA Karangturi, SMAN 82, SMA Immanuel, SMA Saint John, SMA Narada, SMA Kesatuan, SMA Yakobus, and Binus School Bekasi.
- Lead extracurricular programming programs and community education events at BINUS Alsat (ICONATION).
- Deliver regional outreach workshops in Pontianak and Palembang.
- Develop AI learning materials for GreatNusa.
- Deliver generative AI training for small business community development (CommDev UMKM).

## **ACHIEVEMENTS**

- Recipient of the Faculty Development Program (**FDP**) **Scholarship** for master's degree studies.
- Graduated **Summa Cum Laude**, Master of Computer Science, BINUS University.
- Graduated as **Best Graduate**, Bachelor of Computer Science, Soegijapranata Catholic University.
- Graduate of Bangkit Academy 2022, Machine Learning Path by **Google, GoTo, and Traveloka**; led "Kerjamin," the team's capstone project, a community-driven freelancing platform designed to expand income opportunities for workers in rural Indonesia.

## **RESEARCH AND PUBLICATION**

### **JOURNAL ARTICLES**

- **Lighter student engagement recognition in a classroom environment using skeletal keypoints.**

Tarigan, G. A., Elwirehardja, G. N., Nugroho, K. S., & Pardamean, B. — IAENG International Journal of Computer Science, 52(6), 1997 (2025).

- **Skeletal keypoint-based pipeline as a computer vision-based approach.**

Tarigan, G. A., Nugroho, K. S., & Pardamean, B. — Communications in Mathematical Biology and Neuroscience (2026).

### **CONFERENCE PAPERS**

- **Enhancing node classification: Integrating particle swarm optimization with graph convolutional networks.**

Nasari, M., Pradana, R. C., Tarigan, G. A., Masaling, N. A. P., & Tedjasulaksana, J. J. — 2024 International Conference on Intelligent Cybernetics Technology & Applications (ICICyTA), pp. 1239-1243 (2024).

- **Parallelization of LSTM-GRU Architectures for Multivariate Prediction of Stock Prices.**

Tarigan, G. A., Hermawan, E. S., & Girsang, A. S. — 2024 International Conference on Information Management and Technology (ICIMTech), pp. 311-315 (2024).

- **Ablation study: Calibrated adaptive learning ensemble methodology.**

Perdana, G. A., Wijaya, I. I., Fahreza, K. A., & Tarigan, G. A. — Procedia Computer Science, 269, pp. 993-1001 (2025).

- **Indonesian Food Detection with YOLO and Calorie Estimation Using Lightweight LLMs.**

Nasari, M., Tarigan, G. A., Masaling, N. A. P., Joddy, S., & Minor, K. A. — 2025 8th International Seminar on Research of Information Technology and Intelligent Systems (ISRITI), pp. 784-789 (2025).

- **InterQ: A Job Interview Training Platform Using Facial Expression Detection and Answer Scoring.**

Boer, Y., Permatasari, A. C., Kevin, K., Tarigan, G. A., & Suhartono, D. — 2025 8th International Seminar on Research of Information Technology and Intelligent Systems (ISRITI), pp. 841-846 (2025).

- **Analyzing Indonesian political hoax detection system on social media using deep learning and natural language**

processing.

Prastyapradipta, B., Naoko, K., Himawan, R. A. B., Ibrahim, M. A., & Tarigan, G. A. — Procedia Computer Science, 269, pp. 1742-1751 (2025).

- **Computer Vision-Based Pose Estimation for Student Engagement Detection: Trends and Challenges.**

Tarigan, G. A., Elwirehardja, G. N., Nugroho, K. S., & Pardamean, B. — 2025 7th International Conference on Cybernetics and Intelligent System (ICORIS), pp. 1-5 (2025).

- **Product sales recommendation system using item-based collaborative and content-based filtering.**

Harun, J., Yulianto, D., Andrywinata, C., Hermawan, E. S., Pranoto, H., et al. — Procedia Computer Science, 269, pp. 1643-1653 (2025).

## CERTIFICATIONS AND EXPERIENCES

- **SecOps AI-ML Pen Certified Professional**

2025

Practical exam focused on AI/ML security: exploiting LLMs and ML systems via prompt injection, RAG poisoning, and similar attack techniques.

- **Microsoft AI-900**

2025

Microsoft's official certification covering core AI and ML concepts and Azure's AI service stack: cognitive services, computer vision, NLP, and generative AI.

- **Kominfo (Ministry of Communication) Digital Talent Scholarship**

Aug-Sep 2021

Selected for the Digital Talent Scholarship by Indonesia's Ministry of Communication. Completed the Artificial Intelligence track from Huawei, taught at Padjadjaran University, and earned the Huawei Certified ICT Associate in Artificial Intelligence.

## SKILLS

- **AI & Machine Learning**

Python, PyTorch, TensorFlow, scikit-learn, NumPy, pandas, OpenCV

- **AI Assisted Development**

LLM APIs, AI coding assistants/agents, prompt-based code generation

- **Software Engineering & Backend**

FastAPI, REST APIs, SQL, Docker, GitHub Actions, Linux

- **ML Systems & Infrastructure**

MLOps, model deployment, automated evaluation, containerized applications

- **Other**

Java, PHP, C, JavaScript, HTML/CSS

## LANGUAGES

**Bahasa Indonesia:** Native Proficiency; **English:** EF SET: C2 Proficient